

Samantha Blanco

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M.S. Electrical Engineering student at Florida International University with a B.S. in Electronic Engineering and six years of professional experience at Philips across quality, sales operations, process improvement, and technical order execution. Technical interests include electrical design, power systems, energy systems, MEP, hardware validation, field engineering and grid resilience. Experienced in power systems analysis, VLSI design, technical documentation, process improvement, and cross-functional project execution. Available for full-time roles starting September 15, 2026 under post-completion OPT.

EDUCATION

Florida International University

Miami, FL

Master of Science in Electrical Engineering GPA 3.93

Jan. 2025 – Aug 2026

Santa Maria la Antigua Catholic University

Panama

Bachelor's Degree in Electronic Engineering GPA 3.53

May 2015 – Aug 2019

EXPERIENCE

Graduate Student Program Associate

Mar 2026 – Present

Florida International University

- Increased program visibility by generating **12,341 Instagram views** and reaching **516 accounts**, with **66% of views from non-followers**, through targeted social media content promoting scholar achievements and graduate preparation resources.
- Developed and managed the program newsletter, curating academic resources, opportunities, program updates, and student highlights to improve communication and engagement with scholars.
- Supported graduate school readiness by assisting with departmental operations, student communications, workshop logistics, recruitment efforts, and program initiatives designed to support scholar success.

Senior Customer Order Specialist

Jan 2024 – Dec 2024

Philips

- Reduced customer delivery confirmation time by **85%**, from **7 days to 24 hours**, by redesigning the new order process and implementing a scalable order-tracking system.
- Managed end-to-end customer order operations for **Imaging Systems modalities** across direct and indirect sales, ensuring alignment with technical, contractual, and quality requirements.
- Led **Diagnostic X-Ray system** order operations, coordinating workflow execution, issue resolution, and production scheduling with manufacturing plants.
- Created and maintained **SAP Sales Orders**, supporting Purchase Order generation, delivery date assignment, and on-time production planning.

Sales Operations Planner

Apr 2022 – Jan 2024

Philips

- Reduced quotation and order processing time by **35%** by implementing a structured sales operations framework that standardized workflows, clarified responsibilities, and improved execution across teams.
- Increased third-party supplier participation in quotations by **25%** by restructuring the third-party buying process and strengthening coordination between sales, procurement, and operations.
- Reduced operational errors by **68%** by strengthening collaboration between commercial, configuration, and back-office teams throughout the quotation and order intake process

Configuration Product Engineer

Mar 2020 – Apr 2022

Philips

- Configured Philips medical equipment across **multiple healthcare modalities**, including Monitoring Analytics & Therapeutic Care, Respirionics, CT, Ultrasound, X-ray, MRI, and Healthcare Informatics Solutions, by aligning technical specifications with customer, compliance, and business requirements.
- Supported commercial teams across **Latin America** by preparing accurate quotations, assisting negotiations, and providing technical and business guidance throughout the proposal, contract, and purchase order process.
- Improved quotation and order execution quality by applying **Lean, Continuous Improvement, KPI monitoring, and CAPA methodologies** to streamline workflows, reduce inefficiencies, and maintain compliance standards

Quality Engineer

Jan 2018 – Mar 2020

Philips

- Improved quality reporting efficiency by developing internal tracking tools using **advanced Excel and Macros**, increasing accuracy, transparency, and data management within the Service Record Review Team.
- Supported patient and operator safety by conducting risk assessments on medical equipment failures and collaborating with manufacturing teams to define corrective actions based on service and maintenance data.
- Evaluated service events from call center data, field reports, and customer feedback to determine regulatory complaint status, ensuring compliance with Philips' **Quality Management System** and global regulatory requirements.

Information Technology Intern

Aug 2015 – Nov 2016

Philips

- Resolved employee hardware and software issues by troubleshooting equipment, installing applications, deploying IT resources, and managing support tickets.
- Supported digital transformation efforts by maintaining and improving the company archive system, including the digitalization of physical records.
- Assisted IT engineers with daily technology operations, contributing to smoother system maintenance, user support, and internal service delivery.

PROJECTS

SMR-Enabled Resilient Microgrids | *Literature Review, Energy Systems, Grid Resilience* Feb 2026 – Present

- Conducted comparative research on small modular reactors, solar-plus-storage, diesel generation, and natural gas for critical-infrastructure microgrid applications.
- Evaluated technologies using resilience criteria including dispatchability, duration capability, carbon emissions, fuel security, siting, licensing, and cost tradeoffs.
- Synthesized findings into a technical paper/poster focused on the suitability of SMRs for resilient microgrid applications.

Power Systems Analysis | *Power World, Python* Feb 2026 – Apr 2026

- Modeled power system operating scenarios in PowerWorld, including load variations and line contingency cases, to evaluate grid behavior under changing system conditions.
- Analyzed active and reactive power responses across buses and system scenarios using structured data tables and Python-based visualizations.
- Developed technical plots and summaries to interpret voltage response, contingency impacts, and overall system performance trends.

Energy Efficiency Assessment | *Electrical Energy Audit, Consumption Analysis, Sustainability*

- Performed an electrical energy audit for a university building to assess consumption patterns, energy use behavior, and opportunities for improved efficiency.
- Analyzed electricity consumption data to identify inefficiencies and evaluate strategies for reducing energy demand.
- Created an energy-saving guide with recommendations to reduce electrical consumption and promote a more sustainable campus building environment.

VLSI Design | *Cadence Virtuoso* Jan 2025 – Dec 2025

- Designed and analyzed transistor-level circuit layouts using Cadence Virtuoso as part of graduate VLSI design coursework.
- Created schematic and layout views for CMOS-based circuit designs, with attention to layout structure, design flow, and circuit behavior.
- Gained exposure to VLSI design methodology, semiconductor design tools, and layout-level implementation.

TECHNICAL SKILLS

Electrical, Power & Energy:Electrical load calculations, NEC familiarity, power systems fundamentals, power distribution, energy systems, grid resilience, circuit analysis, electronics

Tools & Systems:Cadence Virtuoso, PowerWorld, Python, SAP, MATLAB, Salesforce, SOFON, Microsoft Office

Project & Process Improvement :Six Sigma Yellow Belt, Project coordination, Project management, sales operations, order management, KPI tracking, process mapping, workflow automation, Lean process optimization, root cause analysis

Quality & Compliance: QMS, CAPA, risk assessment, regulatory compliance, continuous improvement